Magno Charge: An Intelligent Auto Disconnect Charger

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Abstract

Magno Charge introduces a revolutionary charging solution that addresses the pervasive issue of overcharging in mobile devices, enhancing safety, energy efficiency, and battery longevity. By leveraging cutting-edge Electromagnetic properties and seamless wireless communication, this system ensures automatic physical disconnection of the charging cable upon reaching the preset battery level. This not only protects the battery but also promotes eco-friendly energy usage. Magno Charge is a pioneering, sustainable, and user-centric alternative to conventional charging technologies.

Keywords: Overcharging, Battery Safety, Energy Efficiency, Magnetic Polarity Control, Automatic Disconnection, User-Centric Design.

I. INTRODUCTION

Background

Mobile phones are a necessity in today's tech-driven environment, which makes safe and effective charging crucial. But conventional chargers frequently aren't able to stop charging when a gadget reaches it full capacity, which causes a number of problems:

- Battery degradation
- Energy Waste
- Hazards to Safety
- Trickle Charging

In order to solve these problems, the Magno Charge has a magnetic mechanism that, once the battery is fully charged, automatically disconnects the charging wire, protecting the battery and improving safety all at the same time.

Problem Statement

The mobile device charging solutions available today do not offer sufficient safeguards against overcharging. The constant flow of electricity after the battery is fully charged can lead to overheating, which speeds up battery ageing and may even cause harm to the gadget. Furthermore, it is inconvenient and frequently forgotten to unplug the charger after the battery is fully charged, which results in trickle charging and a shorter battery life.

Objectives

The Magno Charge project has a set of well-defined objectives driving it to revamp

mobile charging technology while bringing about convenience and safety for users. The two main goals include:

- **Automatic Disconnection:** This is the core idea behind Magno Charge, which should design and develop a charging system that automatically disconnects when it reaches full battery charge. That way, intervention is not needed, and optimal battery health is ensured as well as minimum energy wastage.

- **Battery Protection and Longevity**: One of the major reasons for battery degradation is overcharging, which shortens the general life of portable electronics. Magno Charge would avoid this problem through a smart cut-off mechanism that would prevent excess power input, keeping the batteries fresh and in use longer.

- **User Convenience and Automation:** Traditional chargers require users to manually disconnect the power source when charging is done. However, a lot of people forget to do so, leading to overcharging-related issues. Magno Charge automates the process; it provides the user with hassle-free charging which increases user satisfaction and safety in devices.

- **Enhanced Safety Measures**: Overcharging is a critical safety issue that can lead to overheating, short circuits, and even fire hazards. Magno Charge is designed with integrated safety features that monitor charge levels in real time and prevent excessive power flow, ensuring a safe charging environment for users.

- **Integration with Smart Technology:** The project integrates Magno Charge with wireless communication and smartphone applications, which allow users to monitor charging status, customize disconnection thresholds, and receive real-time notifications. This feature adds a layer of intelligence to the charging process, enabling users to have complete control over their device's power management.

- **Customizable Charging Options**: Different users have different charging requirements based on device type and the usage pattern. Magno Charge gives an option to customize the charging threshold wherein users can set predefined battery levels at which the charger auto-disconnects.

- **Future Scalability and Adaptability**: Magno Charge is designed with future scalability in mind. The technology can be adapted for various electronic devices beyond mobile phones, including laptops, tablets, and electric vehicles. This adaptability ensures long-term relevance and potential expansion into other markets.

Through these objectives, Magno Charge stands out as a revolutionary innovation that transforms the charging of a mobile through the safe, efficient, and user-friendly way it's conducted. Automation, smart technology, and energy efficiency intertwine in this advancement to present the future development in power management and consumer electronics.

II. LITERATURE REVIEW

Current charging technologies focus much on fast charging or wireless charging, without attention to the serious problem of overcharging. Overcharging is well-documented and is known to contribute to loss of battery life and efficiency as well as pose some safety concerns. Many of today's charging solutions have taken the focus away from the smarter disconnection, which stops excessive flow of power after the battery reaches its full capacity.

Several studies have addressed the detrimental impact of continuous charging on lithium-ion

batteries, that are widely used in handheld devices. Accumulated charging cycles are known **to** eventually cause chemical degradations at the battery cell level that result in poor capacity and further shortens life. Long extended charging also wastes energy resources, thus upping electricity and carbon emissions from the usage. While some manufacturers have begun to use adaptive charging techniques where the charging speed is reduced around full capacity, they do not provide an auto-shutoff feature.

Magno Charge addresses these shortcomings by integrating state-of-the-art smart charging technology with magnetic polarity control. Unlike conventional chargers, it actively monitors battery levels and autonomously disconnects the power supply once the battery reaches full charge. This approach not only ensures enhanced battery safety and longevity but also improves energy efficiency by preventing unnecessary power usage. In addition, Magno Charge has wireless monitoring, making it possible to track the charge level and allow users to choose the threshold on disconnection with a smartphone application, thus raising the level of user control and ease.

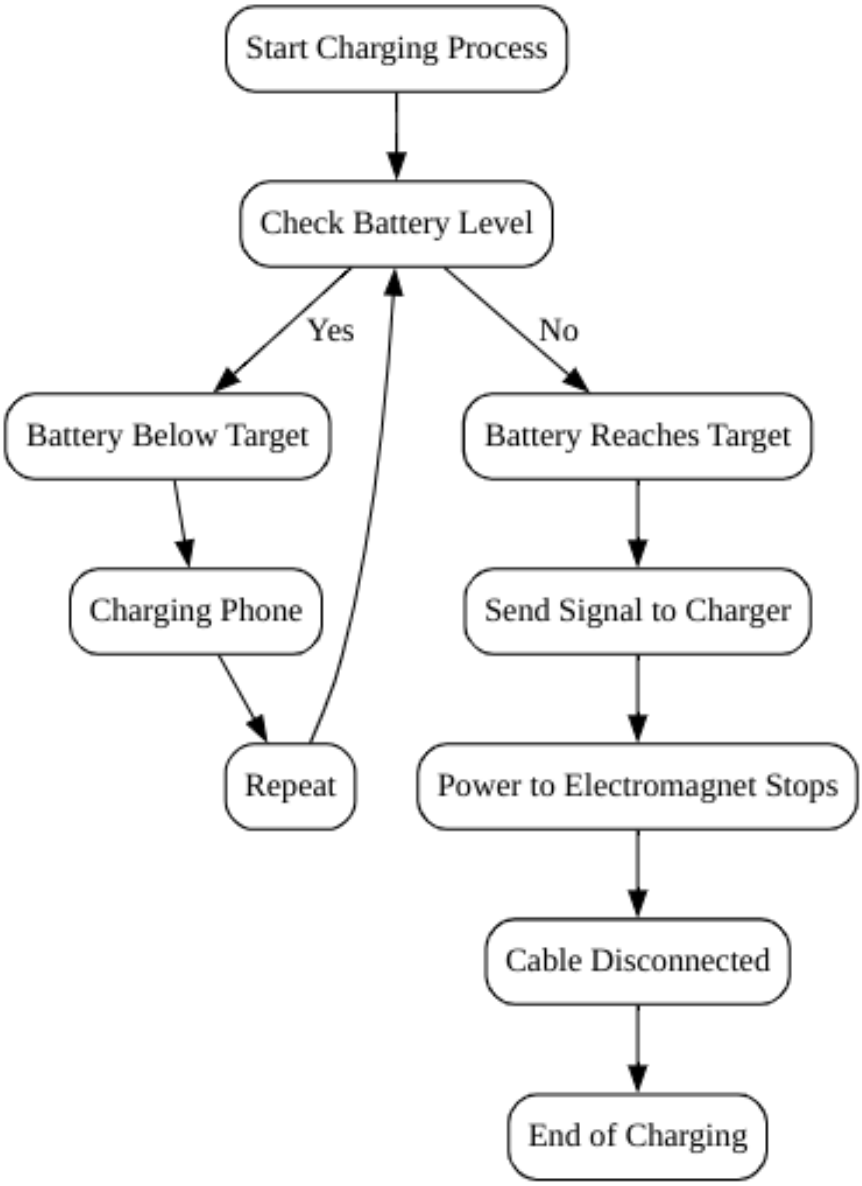
By doing this, the Magno Charge bridges the technological gap while establishing itself in the charging technology landscape as holistic and forward-thinking, combining innovation and practicality into one solution that frees users from worries about overcharge-related issues. With the upsurge demand for sustainable, efficient energy, Magno Charge marks one of the advanced breakthroughs into the smart charging world. Some appliances include basic safety components but do not provide more comprehensive functionalities such as auto-disconnecting. This greatly raises the risk issues associated with faster-than-average battery life consumption, ineffective use of energy, and even probable hazards. On its part, Magno Charge mitigates these weaknesses with a combination of magnetic polarity-controlled, state-of-the-art, smart charging. This innovative approach does not only provide increased battery safety and life but also improves user experience through the automated disconnection process. Magno Charge thus bridges the technological gap as a comprehensive, forward-thinking solution in the landscape of charging technology.

PROPOSED SYSTEM

Technical Features

- **Automatic Disconnection:** The system employs magnetic polarity inversion to disengage the charging cable automatically when the battery reaches full capacity. This prevents overcharging and extends battery life. when the battery reaches full capacity. This prevents overcharging and extends battery life.
- **Wireless Monitoring:** Users can track real-time battery levels, monitor charge history, and set disconnection thresholds via an integrated smartphone app. The app enhances user control and provides alerts when charging is complete.
- **Customizable Settings:** Users can personalize their charging preferences, including setting charge limits and enabling or disabling specific features through the app interface.
- **Seamless Integration:** The charger's design ensures compatibility with a wide range of mobile devices, making it a versatile and user-friendly solution.
- **Enhanced Safety Mechanisms:** The system includes safeguards against overheating and short circuits, ensuring secure operation under all conditions.

Figure.1 Schematic of Proposed System



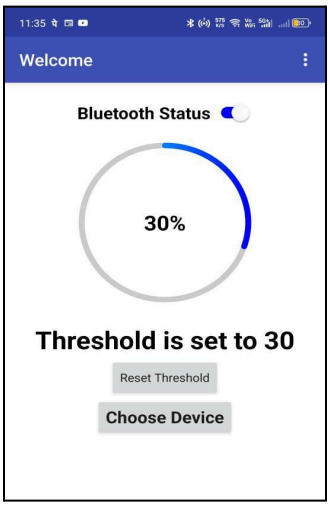
THE PROTOTYPE IN ACTION

The automation of mobile charging disconnection has been a critical focus on energy efficiency and battery health research. Our study explores the design and implementation of a prototype that autonomously terminates charging upon reaching a predefined battery level. At the core of this system is a microcontroller-based junction box that regulates power flow from a conventional mobile charger adapter.

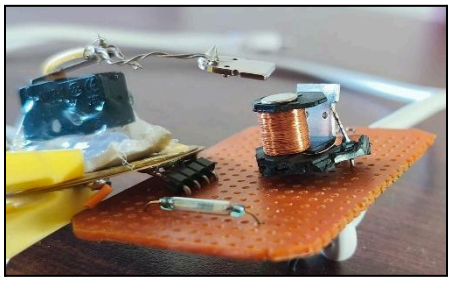
The charger converts AC mains power into DC, which is then supplied to the junction box, integrating an Arduino Nano, a Bluetooth communication module, and an amplifier. This configuration enables precise control and real-time monitoring of the charging process.

A custom-fabricated circuit board within the system houses a solenoid actuator and a magnetic read sensor, both essential for the physical disconnection mechanism. The system employs a metallic flap as the primary switching element, positioned on a stand that ensures clear visibility of the disconnection process. The stand itself incorporates conductor plates to facilitate electrical continuity.

The operational workflow is structured around a mobile application that allows users to set a predefined charge threshold. The microcontroller continuously tracks the device's battery percentage through Bluetooth communication.



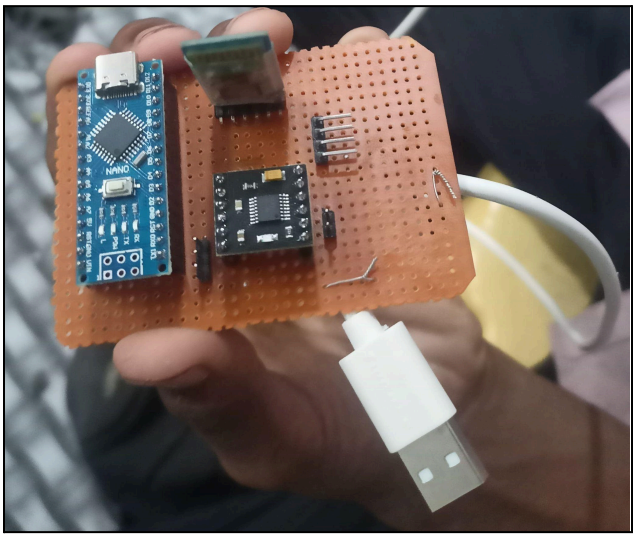
While the charge level remains below the preset limit, the solenoid remains engaged, maintaining current flow through the conductor plates and into the charging cable. Once the threshold is reached, the microcontroller deactivates the solenoid, disengaging the metallic flap and severing the power supply.



Experimental validation of the prototype confirmed the system's ability to automate the charging disconnection process with high accuracy. Real-time monitoring through the mobile interface enhances user control while

preventing overcharging, thereby contributing to improved battery lifespan. Future research will focus on refining solenoid response time, reducing latency in Bluetooth communication, and expanding compatibility with diverse smart devices.

EXPERIMENTAL SETUP AND OUTCOMES



The prototype incorporated a microcontroller based control unit, a Bluetooth module for wireless communication, a solenoid actuator, and a magnetic connector integrated into a USB Type-C charging cable. A metallic flap mechanism enabled controlled detachment upon receiving a signal from the microcontroller. The system was powered using a standard USB power source to ensure compatibility with existing smartphones without requiring modifications to their hardware.

The charging process was monitored in real-time through a custom mobile application that communicated with the microcontroller over Bluetooth. The application allowed users to set a predefined battery threshold, upon reaching which the system would trigger the solenoid to disengage the magnetic connector. The microcontroller periodically retrieved battery status updates, ensuring minimal power consumption while maintaining a continuous monitoring cycle. Upon reaching the defined charge limit, the system successfully executed the disconnection process, effectively demonstrating its capability to halt charging without user intervention.

Observations from multiple trials confirmed that the system reliably detected battery levels and responded accordingly. The detachment mechanism operated smoothly, ensuring that the magnetic connector disengaged as intended when the trigger signal was received.

EXTENSION OF THE PROPOSED SYSTEM

Scalability and Future Growth:

- **Integration with Renewable Energy Sources:** Future editions of Magno Charge can be developed to include renewable energy sources, such as solar panels, thus making the charging process even more eco-friendly.
- **Multi-Device Charging Capability:** The system can be expanded to accommodate several devices at the same time. It can allow a user to charge multiple gadgets while still offering smart disconnection capabilities.
- **Industrial and Commercial Applications:** Beyond consumer electronics, the Magno Charge technology can be applied in industrial and commercial sectors, such as smart grids, electric vehicle charging stations, and battery management systems for large-scale operations.
- **Integration with Smart Home Ecosystems**:

It is also possible to integrate with other smart home ecosystems like Google Home and Amazon Alexa to allow voice-controlled operation and automated scheduling of charging cycles.

- **Cloud-Based Data Storage and Analysis:** Using cloud computing, historical charging data will be stored by Magno Charge, enabling the users to track long-term battery performance and receive predictive maintenance alerts.
- **Portability and Travel-Friendliness:** In the future versions, the product could be optimized for portability and travel to address the requirements of users seeking a reliable source of charging on-the-go.
- **Cross-Compatibility:** Magno Charge shall be capable of charging a multitude of devices with new technologies that may emerge during its lifecycle such as foldable phones, wearables, and Internet of Things.
- **User Feedback and Continuous Improvement:** Establishing a feedback loop with users to gather insights and improve the system based on real-world usage patterns and preferences. This innovative solution has the potential to revolutionize not just mobile charging but the broader landscape of smart energy management by extending the capabilities of Magno Charge beyond its initial scope.

CONCLUSION

Magno Charge is an innovative, practical solution to a common problem with mobile devices related to overcharging. Integration of magnetic polarity control, wireless monitoring, and automatic disconnection enhances the longevity of a battery, reduces energy waste, and improves user convenience. Experimental results have confirmed its effectiveness in preventing overcharging, ensuring thermal safety, and promoting energy efficiency. The appeal of Magno Charge lies in its commercial adoption, marking a milestone in smarter charging technology. It boasts a reliable and sustainable method of managing power.

The adoption of the Magno Charge can also support overall global sustainability because of a decrease in electronic waste and energy consumption. The systems designed ensure that batteries last longer and remain efficient, decreasing their rates of replacement, therefore reducing negative environmental impact. Additionally, seamless integration with smart devices and a user-friendly interface makes it a viable solution for both individual consumers and large-scale applications, such as electric vehicles and industrial battery systems. As the demand for efficient and intelligent power management solutions continues to grow, Magno Charge is poised to become an essential component of next-generation charging technology. The system improves battery longevity, reduces energy waste, and increases user convenience through the integration of magnetic polarity control, wireless monitoring, and automatic disconnection. Experimental results have proven its effectiveness in preventing overcharging, ensuring thermal safety, and promoting energy efficiency. Strong potential for commercial adoption makes Magno Charge a significant advancement in smart charging technology, offering a reliable and sustainable approach to power management.

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